Bidisha Das

Lead AI Engineer | Agentic AI & LLM Platform Engineering | Salesforce Platform
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PROFESSIONAL SUMMARY

Applied AI engineer with nearly 6 years productionizing enterprise **agentic AI**, retrieval, and distributed systems — including 2 years leading architecture and delivery of **LLM-powered agent platforms** at Salesforce. Designs **multi-agent systems** and **Retrieval Augmented Generation (RAG)** pipelines end to end: orchestration, memory and state management, **model routing**, evaluation, governance, and observability. Hands-on with **LangGraph**, **LangChain**, **MCP**, **Python**, **TypeScript**, Java, and **Apex**, delivering **Slack-integrated agentic workflows** on AWS, Docker, and Kubernetes. Security-first practitioner — permission-aware execution, operation allowlists, structured tool contracts, and **prompt-injection hardening**. Delivered AI workflows cutting manual turnaround ∼92% and platform capabilities adopted across 10+ teams; mentors 5+ engineers.

TECHNICAL SKILLS

Agentic AI & Orchestration: LangGraph, LangChain, Model Context Protocol (MCP), multi-agent systems, autonomous agents, agent orchestration, tool/function calling, structured tool contracts, memory & state management, human-in-the-loop (HITL) workflows, LangGraph interrupts with durable state, Claude Code
LLM & Retrieval Engineering: RAG, adaptive & corrective RAG, semantic search, vector retrieval strategies, enterprise knowledge grounding, reranking, chunking & retrieval optimization, structured output extraction, prompt engineering, model routing, model selection & token economics, OpenAI / Anthropic / open-weight APIs; Pinecone, ChromaDB, FAISS, embeddings
Evaluation, Safety & Governance: AI evaluation frameworks, prompt-quality benchmarking, regression / retrieval / routing evaluation, groundedness, hallucination reduction, model performance monitoring, latency & cost analysis, guardrails, prompt-injection defence, input validation, output sanitization, permission-aware execution, operation allowlists, data governance
Languages: Python, TypeScript, JavaScript, Java, Apex, SQL
Salesforce & GTM Stack: Apex, Metadata API, Tooling API, REST/SOAP integrations, queues & sharing configuration, roles and permission models, Salesforce release management & DevOps automation, **Slack apps**, **Slack Bolt SDK**, **LLM-powered Slack agents**, **Slack-integrated agentic workflows**, conversational HITL approvals
Infrastructure, CI/CD & Leadership: AWS (Lambda, API Gateway, ECS), Docker, Kubernetes, Spring Boot, REST, GraphQL, microservices, distributed & event-driven systems, GitHub Actions, Jenkins, CI/CD pipelines, deployment automation, Splunk, observability; system & API design, enterprise AI architecture patterns, architecture & code reviews, root-cause analysis, mentoring

PROFESSIONAL EXPERIENCE

Salesforce **Dec 2024 – Present**
Senior Member of Technical Staff *Bengaluru, India*

- Lead architecture and production delivery of five **enterprise AI agent workflows** for access management, collapsing ∼60 days of manual processing into ∼5 days of self-service execution (**∼92% reduction**).
- Designed a **retrieval-to-action architecture** where natural-language requests retrieve versioned Skills, **enterprise metadata grounds agent planning**, and permission-aware tools execute structured operations; onboarded eight enterprise capabilities to Headless 360 across API exposure, tool discovery, and **MCP execution**, driving adoption and integration standards across **10+ engineering teams**.
- Built **agentic integrations** across roles, groups, queues, sharing configurations, and access policies using **MCP**, REST APIs, Salesforce **Metadata API** and **Tooling API**, Java, Spring, and TypeScript, with schema validation, operation-level authorization, and structured tool contracts.
- Designed and shipped an **LLM-powered Slack agent** in **Python** and **TypeScript** on the **Slack Bolt SDK**, giving stakeholders conversational in-channel access to access-management workflows; interprets natural-language requests, grounds them in enterprise metadata, and executes structured operations with **human-in-the-loop** approval blocks so privileged actions run under the requesting user's own permissions — serving **150+ users across 8 teams** and reducing access-management turnaround from **∼20 minutes to under 3 minutes**.
- Implemented **Apex** automation for programmatic **queue creation** and membership assignment, exposed to agents as a permission-checked tool with schema-validated inputs.
- **Retired five legacy retrieval-driven integrations**, replacing them with a redesigned agent platform on modern agentic, event-driven patterns; consolidated 20+ prompts, permission models, tool definitions, execution paths, and structured-output contracts.
- Implemented an **AI evaluation framework** across retrieval quality, tool selection, permission enforcement, structured outputs, and failure paths, using representative cases and replayed production failures for **regression detection and model performance monitoring**.

- Defined **error-handling, fallback, and graceful-degradation patterns** for non-deterministic agent behavior, keeping authorization and execution decisions deterministic in application code rather than model output.
- Championed **security-first AI engineering**: governed REST and metadata integrations with schema validation, **input validation**, and operation allowlists, cutting misconfigured access-management requests ∼**35%**.
- Deployed and operated agent services on **AWS**, Docker, Kubernetes, and **CI/CD** with Splunk observability and event-driven infrastructure — throughput up ∼**40%** with stronger fault isolation.
- Partner with product, architecture, platform, and security stakeholders to scope AI use cases and evaluate **build-vs-buy** tradeoffs; **mentor 5+ engineers** through design and code reviews.

Disney+ Hotstar **Jan 2023 – Dec 2024**
Software Development Engineer II *Remote*

- Architected high-scale streaming experiences using React, Next.js, JavaScript, and Shaka Player across web and Living Room platforms, serving latency-sensitive, high-availability traffic.
- Reduced player failure rate ∼**70%** with a runtime mechanism that identified and dynamically blacklisted unreliable decoding layers — graceful degradation under real-world failure conditions.
- **Contributed to Shaka Player (open source)**, resolving timestamp-handling defects that eliminated stream repetition during World Cup playback; built Web Worker-based playback reporting for real-time error collection and faster incident investigation.

MakeMyTrip **Jun 2022 – Jan 2023**
Senior Software Engineer *Bengaluru, India*

- Delivered 3+ production workflows across loans, credit, and foreign exchange, translating financial and regulatory requirements into reliable customer experiences; cut application drop-off ∼**25%** and API latency ∼**30%**.
- Integrated 5+ lending, credit-scoring, and forex APIs with standardized retry, validation, and error handling (transaction success +**18%**); embedded KYC and RBI foreign-exchange controls into workflow logic, reducing manual review ∼**40%**.

AutoRABIT **Sep 2020 – Jun 2022**
Software Developer *Hyderabad, India*

- Built reusable platform components, internal dashboards, developer tooling, and automated test infrastructure inside **Salesforce release-management and CI/CD workflows**, improving release reliability ∼**30%**.

SELECTED AI PROJECTS

ResolveFlow — Multi-Agent GitHub Issue Diagnosis with Safe Agentic Execution

- Built a **LangGraph** engineering agent that diagnoses GitHub issues from repository context, closed issues, pull requests, code evidence, and CI signals, over a **RAG pipeline** of ∼2,750 closed-issue chunks with measured relevance thresholds gating whether retrieved evidence may influence downstream reasoning.
- Designed **routing** across deterministic automation, AI-assisted investigation, escalation, and human-approved writes, separating model reasoning from authorization; implemented independent **LLM critique** while keeping approve/escalate decisions deterministic in application code.
- Implemented **LangGraph interrupts with state persisted in Postgres** for durable pause/resume HITL approval, plus GitHub OAuth so approved writes execute under the authenticated user's permissions; write functions refuse execution without valid authorization.
- Built a **50-scenario evaluation suite** (normal, ambiguous, adversarial, failure-path) benchmarking OpenAI, Anthropic, and open-weight models on routing accuracy, behavior, and cost — **86% routing accuracy** at **24% lower execution cost** than baseline.

AI Bug Triage Skill — Salesforce Hackathon

- Built a Claude Code workflow correlating Splunk logs, repository context, source changes, and work-management data into structured investigation reports with evidence, likely failure paths, and root-cause hypotheses — cutting diagnostic-context collection from ∼90 minutes to ∼**15 minutes**.

AdaptiveRAG Evaluation Platform

- Built a Python/AWS platform (Lambda, API Gateway, FAISS, React, Claude) comparing naive, reranked, corrective, and adaptive retrieval with **query-classification routing** over a 100-question evaluation set — **+12% groundedness**, **−21% token cost**, **−18% median latency** versus a fixed baseline.

EDUCATION

St. Thomas' College of Engineering & Technology (MAKAUT) **2020**
B.Tech, Computer Science & Engineering *West Bengal, India*